A Retrieval-Augmented Generation System for a Legal Advisory System in Thai Consumer Protection

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Abstract— When consumers purchase products or services online in Thailand, are often taken advantage of by sellers and need advice on Thai Consumer Protection Laws. However, finding legal experts is expensive and time-consuming, this study presents an intelligent legal system that provides initial advice on Thai Consumer Protection Laws as a guideline for customer. This system leverages the capabilities of Generative Artificial Intelligence (GenAI) to provide users with answers to legal questions. Retrieval Augmented Generation (RAG) are used together to extract and create legal information. This study proposed a new method Semantic Legal Preprocessing (SLP) for preparing a semantic dataset. As a result, the system`s performance was evaluated using standard information retrieval metrics. The Mean Reciprocal Rank (MRR) was found to be 0.54, while both Hit Rate and Precision were measured at 0.71.

Keywords—Generative Artificial Intelligence (GenAI), Large Language Model (LLM), Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Small Language Model (SLM), Legal

I. INTRODUCTION

The online trading market in Thailand has experienced significant growth, transforming consumer behavior and purchasing patterns. The retail and wholesale sector, which includes online sales, has become the largest contributor to the country`s Gross Domestic Product (GDP), shown in Fig. 1. This sector`s value grew from approximately 2,500 billion baht in 2021 to over 2,800 billion baht in 2023, representing a growth rate of roughly 12% over 3 years.

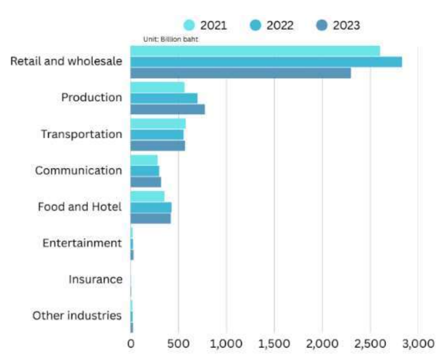

Fig. 1 Overview of the value of e-commerce in 2021-2023 [1].

However, the rapid expansion of the retail and wholesale sector has led to a corresponding increase in consumer complaints. A substantial number of consumers have been exploited by online sellers, forcing them to seek legal assistance. Despite this growing need, accessing legal expertise remains a challenge due to high costs and the its specialized nature. This limited access to legal support creates a critical gap, leaving many consumers vulnerable.

This research focuses on the study and development of a system designed to answer questions related to the Thai Consumer Protection Act (CPA). The CPA is a key piece of legislation that provides rights and protections for consumers in Thailand, particularly against unfair practices in the online marketplace. To develop this system, the specialized legal dataset were manually collected and prepared. This was achieved by leveraging a Large Language Model (LLM) application and applying Prompt Engineering [2] to structure and refine the data effectively.

The core of our system is a language model integrated with Retrieval-Augmented Generation (RAG) and a Small Language Model (SLM) to provide accurate and relevant answers. RAG is a powerful technique that enhances the generative capabilities of language models by allowing them to retrieve information from an external knowledge base before generating a response. This approach ensures that the answers are grounded in specific, verifiable legal documents rather than relying solely on the model`s pre-trained knowledge.

The remainder of this paper is organized as follows: Section 2 reviews related works concerning the application of AI in the legal domain. Section 3 presents our research methodology, the Retrieval-Augmented Generation (RAG) technique, and the evaluation metrics used. Section 4 provide the experimental and results. Finally, Section 5 concludes the paper by summarizing our findings, highlighting the contributions of this research, and outlining future work.

II. RELATED WORK

The emergence of Generative Artificial Intelligence (GenAI) has created numerous opportunities and possibilities across various domains. Its core capability lies in its ability to create new, original content such as text, images, videos, and audio from existing data. A prime example is the use of a text prompt to generate the multimedia contents such as, a complex musical score, a realistic image, and a detailed article. This transformative technology is being applied widely, from creative industries to scientific research.

The Large Language Model (LLM), which is specifically designed to generate new text from vast amounts of existing data. The prominent LLMs that have gained significant attention include GPT-5 from OpenAI [3], LLaMA from Meta [4], and Gemini from Google [5]. These models have enabled the development of powerful applications benefiting a wide range of domains, including the legal field.

The application of LLMs, especially in specialized domains like law [6],[12] and medicine [13], or with sensitive internal corporate data, still faces significant challenges. A primary concern is the phenomenon of hallucination [14], where the model generates information that is factually incorrect, nonsensical, or entirely fabricated. This unreliability poses a serious risk when accuracy is paramount, as a hallucinated answer in a legal context could lead to misinterpretations or incorrect guidance, while in a medical context, it could potentially result in a life-threatening misdiagnosis or improper treatment.

One potential solution to this problem is fine-tuning, which involves retraining a language model on a new, specific dataset. While this can improve performance, it is often a time-consuming and resource-intensive process. Moreover, the model may still suffer from hallucinations when encountering data it has not been specifically trained on.

A more efficient and widely adopted approach to mitigate the hallucination problem is Retrieval Augmented Generation (RAG) [7]. This technique enhances a language model`s ability to provide accurate and relevant answers by allowing it to retrieve information from an external knowledge base before generating a response. Instead of relying solely on its pre-trained knowledge, the LLM uses the retrieved data as a direct reference, which grounds its output in facts and significantly reduces the risk of generating false or misleading information. This method is particularly useful in specialized fields where accuracy is paramount, as it ensures the model's responses are based on up-to-date and domain-specific knowledge.

Several research articles propose different approaches to optimize Retrieval Augmented Generation (RAG) systems [6] as shown in Table I. For specialized fields such as law [7], [9], [12], [10], [11], and customs procedures [8]. These approaches aim to address key limitations of Large Language Models (LLMs), such as hallucinations and outdated knowledge, by grounding them in external, field-specific data. These systems perform well, showing significant improvements in response accuracy and context awareness.

However, several gaps remain. In this study specifically address the sensitive legal domain of consumer protection law and support the Thai language, a context often overlooked in existing research. Our dataset is meticulously curated using prompt engineering techniques to generate structured questions and answers. To ensure a robust evaluation, we also employ appropriate metrics for RAG, such as MRR, Precision, and Hit-Rate, which provide a more comprehensive assessment than traditional methods.

III. Proposed Method

This section describes the methodology used in this study, which explains the detailed steps and instruments used in conducting this study, divided into the following steps.

A. *Retrieval-Augmented Generation (RAG)*

In this study, we have developed a GenAI system for answering questions on Thai consumer protection law using RAG technique. We chose RAG over direct fine-tuning of a Large Language Model (LLM) because it effectively mitigates the major problem of hallucination a significant issue when dealing with specialized knowledge like law. RAG enhances the LLM`s ability to generate

TABLE I. THE RELATED WORKS COMPARISON

No.	Ref./Year	Objectives	Techniques	Dataset	Results
1	[6], 2025	To enhance RAG performance by resolving the stylistic mismatch between user queries and document content	Hypothetical Prompt Embeddings (HyPE)	◦ MS MARCO ◦ Ragas-WikiQA ◦ RAG-dataset-12000 ◦ MultiHopRAG	HyPE can improve retrieval context precision by up to 42 % and claim recall by up to 45 % compared to standard approaches
2	[7], 2025	To provide a comprehensive overview of RAG methods, challenges, and applications in the legal domain.	RAG Stages, The core processes include Information Retrieval (IR), Augmentation, and Generation	◦ BorderLegal-QA ◦ JEC-QA ◦ CJRC ◦ CAIL2020 ◦ CAIL2021	improve LLM accuracy and reduce hallucinations in critical fields like law
3	[8], 2025	To propose an Intelligent Customs Clearance Assistant (ICCA-RAG) for customs clearance using RAG to manage multimodal data and resolve ambiguous queries.	Multimodal Parsing, Processes PDFs, images, and tables using tools like PyMuPDF and Tesseract OCR.	dataset of Chinese customs materials, including tariff chapters, customs laws, and inspection and quarantine ordinances	Answer correctness rose by 20.1%, relevancy by 15.3%, and faithfulness by 18.7%,
4	[9], 2025	To propose a new framework using recursive feedback to improve transparency and prevent hallucinations and bias in legal applications.	Hybrid Fine-Tuned Generative LLM (HFM)	◦ Dsynthetic ◦ Domain-specific Q&A	24% performance gain over general LLM improved by 9% on general tasks and 38% on legal domain-specific tasks compared to the baseline LLaMA-3-8B
5	[10], 2025	To optimize legal text summarization by integrating a dynamic RAG system with domain-specific adaptation.	Dynamic Legal RAG System, Legal (NER)	◦ Indian Penal Code ◦ Civil Procedure Code ◦ Criminal Procedure Code	BERTScore 0.89 ROUGE-L 0.8894 Cosine similarity 0.94
6	[11], 2023	To propose a simple, zero-shot prompting method to teach LLMs to predict legal judgments.	Legal Syllogism Prompting (LoT)	◦ CAIL2018 ◦ Criminal case	Logic-of-Thought (LoT) method achieved an accuracy of 0.6850

accurate and trustworthy responses by grounding them in factual, external data. The general RAG system operates through the following key components shown in Fig. 2.

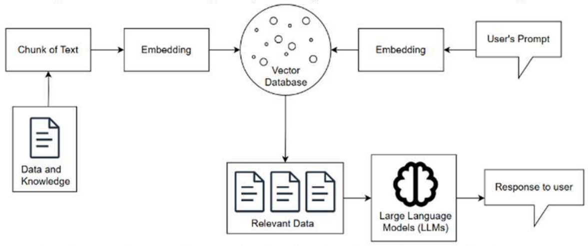

Fig. 2. Retrieval Augmented Generation (RAG)

First, the data or knowledge base is broken down into smaller, manageable pieces called "chunks of text." This process is essential for efficient information retrieval. Each text chunk is converted into numerical representations called "embeddings". A specialized language model performs this conversion, ensuring that the vectors capture the meaning and context of the original text. The closer the vectors are in the vector space, the more semantically similar the texts are.

These semantic vectors are stored in a Vector Database, which is a specialized database designed to store and quickly search for vectors based on their similarity. When a user submits a question, the system will convert a question into vectors using the same model, then system uses the question`s vector to perform a search within the Vector Database. This process, known as "Dense Retrieval", finds the chunks of text that are most semantically similar to the user`s question. Unlike traditional keyword-based searches, Dense Retrieval understands the intent and meaning behind the query. The most relevant chunks are then retrieved as "Relevant Data". Finally, the Small Language Model (SLM) receives two key inputs, the user`s original question and the relevant data retrieved from the Vector Database. Using this retrieved information as a factual reference, the LLM generates a coherent, accurate, and natural-sounding response to the user.

B. Dataset Collection

The dataset for this study was carefully collected and curated to ensure comprehensive coverage of Thai consumer protection law. Data were sourced from official and reliable channels, including government agencies and public interest organizations. The collection process can be divided into two primary sources shown in Table. II.

1. Official Government Sources: The main legal documents were collected from the official website of the Office of the Consumer Protection Board (OCPB), a key government agency responsible for enforcing the Thai Consumer Protection Act (CPA) 1979. We utilized 11 PDF files containing the primary laws, sub-regulations, and ministerial announcements officially published in Thailand. These documents form the foundational knowledge base of our legal system, ensuring all answers are grounded in formal statutes.

2. Articles from Public Interest Organizations: To supplement the official legal texts, we gathered a large number of articles, news, and case studies from the website of the Consumer Council of Thailand. This organization provides valuable real-world context and practical interpretations of consumer protection laws through its extensive collection of articles and public-facing documents. Approximately 47 articles were collected to provide a more comprehensive understanding of how the law is applied in practice.

TABLE II. SUMMARY OF DATASETS USED FOR RAG.

Office of the Consumer Protection Board (OCPB)		
Act Title (Latest Update)	Year of Enactment	Number of Sec.
Drug Act.	1967	129
Consumer Protection Act.	1979	63
Food Act.	1979	78
Hazardous Substances Act.	1992	93
Road Accident Victims Protection Act.	1992	47
Direct Sales and Direct Marketing Act.	2002	56
Consumer Case Procedure Act.	2008	66
Product Liability Act.	2008	16
Cosmetics Act.	2015	94
Personal Data Protection Act.	2019	96
Mediation Act.	2019	72

Consumer Council of Thailand	
Content Type	Number of Content
Article	20
News	12
Cast Studies	15

C. Data Preprocessing

Before to applying the Semantic Legal Preprocessing (SLP) method, we meticulously prepared the raw data to ensure its quality and structure. The legal documents and articles collected, originally in PDF format, were first converted into plain text files. This step was crucial for enabling our automated processing pipeline.

For processed legal documents, such as the Thai Consumer Protection Act (CPA) of 1979, by dividing them into logical sections like chapters and articles. This ensures each file contains a complete legal concept, preserving the original context. Similarly, articles and case studies from the Consumer Council of Thailand were saved as individual text files, each representing a single topic. This preprocessing resulted in a well-organized dataset of 2,000 rows of legal information. These files are now ready for the Semantic Legal Preprocessing (SLP) stage, where they will be prepared for our RAG system to ensure accurate vector creation and retrieval.

D. Semantic Legal Preprocessing

A novel method, Semantic Legal Preprocessing (SLP), is proposed. to enhance the performance of a RAG system for the legal domain. The core benefit of SLP is its ability to transform complex, unstructured legal documents into a semantically rich and easily understandable format. This process significantly improves the quality of the knowledge base, ensuring that the system's generated answers are more accurate, relevant, and reliable. SLP addresses the limitations of traditional RAG by focusing on meaningful data segmentation, which is a crucial step for handling the intricate nature of legal information, as shown in Fig. 3.

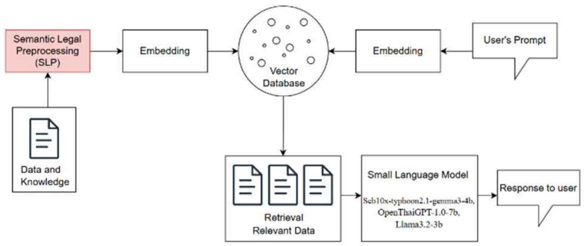

Fig. 3. Our proposed method (RAG with SLP)

After gathering and preparing data from various sources, we apply the SLP method, which differs from basic RAG. In traditional RAG, documents are often split into arbitrary chunks based on a fixed number of characters or tokens, a process that frequently lacks semantic meaning. Such basic chunking is not suitable for complex legal contexts that require deep reasoning. SLP replaces this problematic step with a more meaningful segmentation that enhances understanding. Prompt Engineering was used with the ChatGPT-4o Model to transform these legal documents into a structured format of Question (Q), Answer (A), Explain (E), and Sample (S) pairs, as shown in Fig. 4. This output is easier to read and easy to understand.

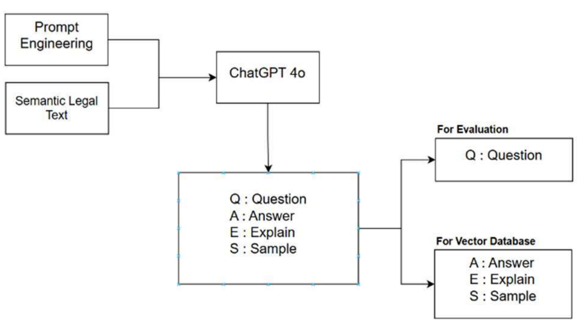

Fig. 4. Semantic Legal Preprocessing (SLP)

E. Embedding

For the embedding models used in this study, several prominent was selected as follows:

BGE-M3, a text embedding model that supports multiple languages, has a vector size of 1024, and can process data ranging from short sentences to sentences with a length of 8192 tokens. The model also supports multi-functionality in retrieval, including dense retrieval, multi-vector retrieval, and sparse retrieval. In this study, we focus on the use of dense retrieval, which is a single embedding, using the same encoder to embed text on both the vector database and the query side, resulting in fast retrieval and saving processing power. In addition, there is another outstanding model with good retrieval performance.

WangchanX-Legal-ThaiCCL-Retriever, which is a fine-tuned model from bge-m3, focusing on business and commercial law data. It is designed for use in Retrieval-Augmented Generation (RAG) systems. The data obtained from Semantic Legal Preprocessing (SLP) is then embedded in a single embedding text format and converted to a JSON string for saving in the TiDB database. When a query is sent to the system, the embedding text is converted to a JSON string and the cosine distance between the query and the five closest retrieved documents from the database is measured.

F. Vector Database

Vector databases are a specific type of database designed to store and manage vector embeddings, which are numerical representations that meaningful. Unlike traditional databases that store structured data in rows and columns, vector databases are optimized for handling these high-dimensional vectors, making them ideal for tasks involving similarity search. To store the data for this system, uses TiDB Cloud, a standard SQL-based, scalable, and automated database system suitable for large volumes of data and privacy. This database was used to store strings converted from vectors and documents for retrieval.

G. Evaluation

Evaluation for Retrieval-Augmented Generation (RAG) systems, after the relevant information retrieval step before sending to Small Language Model (SLM) in a query format. For evaluating the performance in this study, there are 3 evaluation metrics used such as MRR, Precision@k, and Hit rate.

1) Mean Reciprocal Rank (MRR): Used to measure how "correct" the answer is in the order of the retrieval returned by the system. If the result is in the top of the k-number of results, it will be calculated the rank of answer match with ground truth that is correct answer and only first one matching. The average reciprocal rank of all questions yields the MRR. A high MRR means that correct answers are often at the top of the list.

$$MRR = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{rank\_i} \tag{1}$$

2) Precision@k: Used to measure the “accuracy” of a retrieval system on the top-k results by looking at how many of the top k results are actually relevant to the question. Relevant items represent documents or answers that match ground truth that is correct answer.

$$\text{Precision@}k = \frac{\text{relevant items in Top } k}{k} \tag{2}$$

3) Hit Rate@k: It is the proportion of all questions for which the system can “find” at least one truly relevant document or answer within the Top-k results. Queries + relevant in Top-k represent how many queries relevant to an answer. This metric evaluates the correct answer of a system retrieval. If there is a correct answer regardless of the rank or number, it is considered a hit.

$$\text{Hit Rate@}k = \frac{\text{queries} + \text{relevant in Top } k}{\text{all queries}} \qquad (3)$$

With 3 metrics used to evaluate the retrieval accuracy in this study, the efficiency of this process can be evaluated.

H. Hardware and Software

The experimental setup for this study consisted of a system with an Intel Core i5-12500H (2.50 GHz) processor (3.33 GHz), 16.00 GB of RAM, and a Nvidia GeForce RTX 3050 (4 GB) graphics card. The software configuration included Python (3.12.8) and the Visual Studio Code (VS Code) editor for developing the core system. TiDB was utilized as a private, cloud-based vector database for the scalable and efficient storage of semantic legal data. A prototype of the user interface was developed using Gradio, which provided a user-friendly platform for testing the system.

IV. Experiment And Results

In this section, presents the experimental results that have been conducted. The experiments were conducted with the 6 embedding models for converting text to numeric vectors. The random selected of 25% of the questions from the available data in the database for each model to measure the accuracy of answering legal questions, after that applied the best embedding models with 3 Small Language Models (SLMs), such as Llama3.2-3b, Scb10x-typhoon2.1-gemma3-4b, and OpenThaiGPT-1.0-7b, for generation and evaluate with Faithfulness matric.

A. Result

There are 3 accuracy metrics has evaluated [9], [15] with 6 selected embedded models and obtained the results as shown in Table III.

To evaluate the performance of all three models with the 6 selected embedded models [16], and the results are shown in Table III. It can be seen that even though all 6 embedded models support Thai language such as BGE-M3, Thai CCL, ARCTIC, E5 Base, MiniLMV2 and MRL. Mean Reciprocal Rank (MRR) retrieval sequence is significantly high, the BGE-M3 embedded model archive 0.54 for MRR. The results shown that the system has efficient retrieval and can retrieve correct answers at the initial level.

Table III. EXPERIMENT RESULTS OF RETRIEVAL

Embedding Models	Evaluation Metric		
	MRR	Hit Rate	Precision
BGE-M3	**0.54**	0.70	0.70
Thai CCL	0.52	**0.71**	**0.71**
ARCTIC	0.49	0.64	0. 64
E5 Base	0.47	0.66	0.66
MiniLMV2	0.28	0.42	0.42
MRL	0.01	0.00	0.01

In terms of Hit Rate and Precision, WangchanX-Legal-ThaiCCL-Retriever has a high value of 0.71. With these three indicators, which are important indicators for the RAG system. This embedding model will be used as vector database. And then Llama3.2-3b, Scb10x-typhoon2.1-gemma3-4b, and OpenThaiGPT-1.0-7b models were used as a base models to transform data that retrieval from vector database as a prompt and then generate response to user.

Faithfulness matric used for evaluation 3 generation models with random 100 questions to measures if the generated answer is factually supported by and consistent with the provided source context. Scb10x-typhoon2.1-gemma3-4b has high value of 0.1249 as shown in Table IV indicate ensuring it avoids hallucination.

Table IV EXPERIMENT RESULTS OF GENERATION

Generation Models	Faithfulness
Llama3.2-3b	0.1128
Scb10x-typhoon2.1-gemma3-4b	**0.1249**
OpenThaiGPT-1.0-7b	0.1179

As shown in Fig. 5. The software prototype was developed by Gradio which present the result with complete answer as structured message, including answer of legal question and summary. This result has meaningful and easy to understand.

V. Conclusion

The experimental results demonstrate that the performance of a Thai legal information retrieval system is highly dependent on the quality of its data representation. A retrieval system that utilizes a dedicated data embedding model, trained specifically on Thai legal data, outperforms systems that rely solely on simple text chunking and vectorization, which often lack precision and fail to capture semantic meaning effectively. The effectiveness of the data embedding model is clearly evidenced by key RAG evaluation metrics: the Mean Reciprocal Rank (MRR), Hit Rate, and Precision. Our findings show that the Thai WangchanX-Legal-ThaiCCL-Retriever model, specifically fine-tuned for this domain, achieved an accuracy of up to 0.71 in both precision and Hit Rate. This validates that a specialized, domain-adapted approach is crucial for achieving accurate and efficient information retrieval in Thai legal contexts.

Regarding the generative component, all tested Small Language Models (SLMs), Scb10x-typhoon2.1-gemma3-4b provided promising results for legal tasks, as shown in Table IV, and Fig. 5. This is a significant finding as it suggests that high-quality legal information can be generated without the need for large, computationally expensive models, thereby conserving energy and enabling deployment on devices with smaller processors.

A key limitation of this study is the constrained scope of the legal domain addressed and limited dataset. While the findings are robust for consumer protection law, the applicability and performance of the model in other complex legal areas remain unexplored. Additionally, the reliance on a single, self-contained dataset, while useful for controlling variables, may not fully capture the breadth and diversity of real-world legal queries and documents. However, the capabilities of RAG and SLP, can generate additional datasets in the future and further enhance the robustness of the system.

The methodology developed in this research offers a flexible and powerful framework that can be extended to other legal domains. Future work should focus on applying and validating this approach with datasets from other legal fields, such as the Civil and Commercial Code or various business laws. This would not only expand the practical utility of the system but also provide a more comprehensive understanding of its generalizability and limitations across different legal sub-domains.

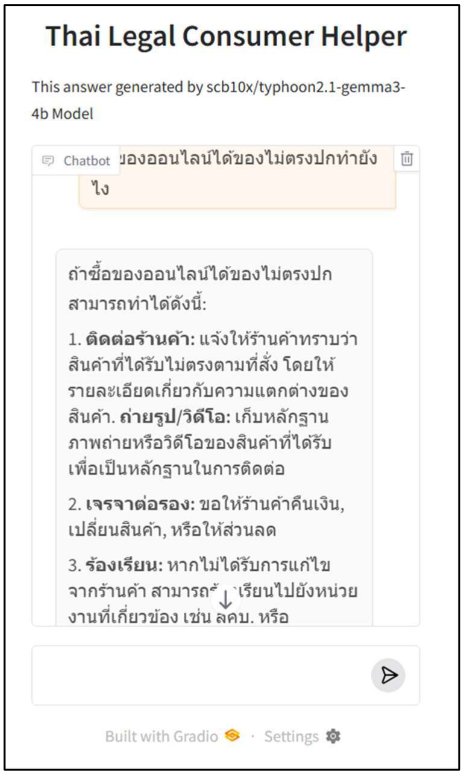

Fig. 5 The result of Thai Legal Consumer Helper Prototype.

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